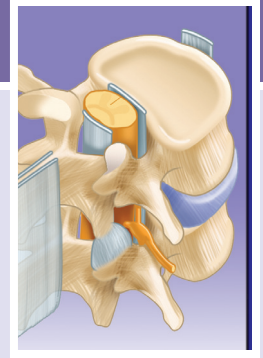


# Stingers

*Sandip P. Tarpada, Woojin Cho*



## INTRODUCTION

The terms “*stinger*” and “*burner*” are synonymous, and together represent a range of brachial nerve plexopathies. Initially described by Chrisman et al.,<sup>1</sup> stingers are almost invariably trauma-mediated, unilateral, reversible injuries of the upper extremity. While stingers are among the most commonly cited injuries of the cervical region, they are solely pathology of peripheral nerves. Most often, traction secondary to trauma will cause burning and paresthesia along a C5–C6 motor and sensory distribution, resulting in transient weakness and numbness.<sup>2–4</sup> The vast majority of cases resolve within minutes. However, in rare cases, symptoms may persist for weeks to months, resulting in significant impairment to quality of life.<sup>4</sup>

Contact sports, particularly American football and rugby, are among the most commonly cited settings for stingers. Indeed, over half of all contact-sport athletes will develop a stinger during their career. While previous studies have found that up to 65% of football players report at least one stinger injury, recent work suggests a lower incidence of about 20% to 30%.<sup>5,6</sup> Defensive ends/linebackers and offensive linemen appear to be at the highest risk. Similar estimates have been made among rugby players and wrestlers.<sup>5–7</sup> However, some have called into question the accuracy of these figures, as stingers have traditionally been underreported. Recurrence is common, accounting for nearly one in five stinger injuries.<sup>6</sup> Recurrent injury increases the risk of permanent nerve damage, which underscores the importance of proper athlete education and injury prevention. The purpose of this chapter is to review the history, exam, imaging, decision, and treatment options for the management of stingers. The authors will also discuss their preferred approach, return to play (RTP), and future considerations.

## HISTORY

The most common setting of a stinger injury in the athlete is during play. Compressive trauma or traction occurs to an athlete whose neck is rotated contralateral to the blow. Such trauma may also depress the ipsilateral shoulder, increasing brachial plexus strain. Depending on the extent of injury, the patient may instinctually “shake out” the affected extremity, support it with the unaffected contralateral extremity, or report generalized pain in the area of the supraclavicular fossa.<sup>8</sup> Compression, traction, and hyperextension injury are thus the three major

mechanisms of stinger injury, and will be discussed with greater detail in a further section.

Pain from a stinger injury, true to its namesake, is frequently described as a burning paresthesia, with associated numbness and heaviness of the affected extremity. While the C5–C6 distributions are most commonly involved, circumferential rather than dermatomal radiation of pain down the extremity is not uncommon. Sensorimotor deficits, when present, typically rectify within several minutes. However, fewer than 1 in 10 patients may have residual symptoms lasting several weeks<sup>3</sup>

## PHYSICAL EXAMINATION

A physical examination is warranted in any athlete with a suspected injury to the brachial plexus, as it can provide crucial details for further management. Evaluation initially begins with a focused cervical spine exam, followed by the characterization of musculoskeletal or neurologic abnormalities. Specific clinical exam techniques may then be used to identify the lesion location.

In the setting of apparent trauma to the neck, loss of consciousness, or gross deformity, the patient’s cervical spine should be empirically immobilized until further imaging can be performed. Additionally, tenderness to direct palpation of the cervical spine in the presence of any trauma, even in the absence of obvious trauma, warrants application of a hard cervical collar. After the exclusion of cervical instability, basic evaluation of neck and extremity range of motion (ROM) and sensation should follow. It is important to note that stinger injuries are rarely bilateral, and do not affect the lower extremities; such signs point towards a more sinister pathology of the cervical spine.

Sensorimotor anomalies may be quickly and efficiently screened for on the field. Weakness of the deltoid and diminished wrist extension suggests C5 and C6 motor deficits, respectively.<sup>9</sup> These findings, along with a reduction in lateral upper extremity and forearm sensation, represent the most common presentation of stinger injuries. Triceps weakness and reduced wrist flexion suggest C7 pathology, while weakness in finger flexion or medial forearm sensation suggests involvement of C8. A T1 abnormality, though rare, presents with loss or diminished sensation to the medial arm, and weakness of the intrinsic muscles of the hand (Fig. 127.1 and Table 127.1).

Evaluation of the native anatomy of the patient may also provide clinical information. The upper trunk of the brachial plexus runs superficially approximately 2 cm above the clavicle

## Abstract

Stinger injuries are common among athletes participating in contact sports. They typically present with transient weakness, numbness, or paresthesia following direct trauma, compression, or traction to the brachial plexus. Symptoms last from seconds to minutes, and rarely persist beyond the day of injury. Concurrent cervical spine injury, while less common, poses a serious risk to injured athletes and must be ruled out. In general, this requires comprehensive evaluation by a certified athletic trainer at field side, as well as appropriate medical follow-up. Return to sport is quite good, and should be reserved for patients whose symptoms have resolved, and subjectively feel comfortable with resuming contact activity. The most common complication of a stinger is recurrence, which occurs in over half of all affected athletes. Awareness of the stinger injury among health care providers is crucial, as diagnosis is commonly missed. Protective equipment, such as the Cowboy collar, the Kerr collar, and the cervical neck roll, may be used to prevent stinger occurrence.

## Keywords

stingers  
burners  
football injuries  
wrestling injuries  
brachial plexopathy  
neuropraxia

at the level of the C6 vertebral body. Known as the Erb point, this location describes the convergence of the C5 and C6 nerve roots. Direct palpation of the Erb point may elicit a Tinel sign—reproduction of numbness and tingling in the distal extremity along a C5–C6 distribution.<sup>9,10</sup> The Tinel sign has both variable sensitivity (45% to 75%) and specificity (55% to 75%).<sup>10,11</sup>

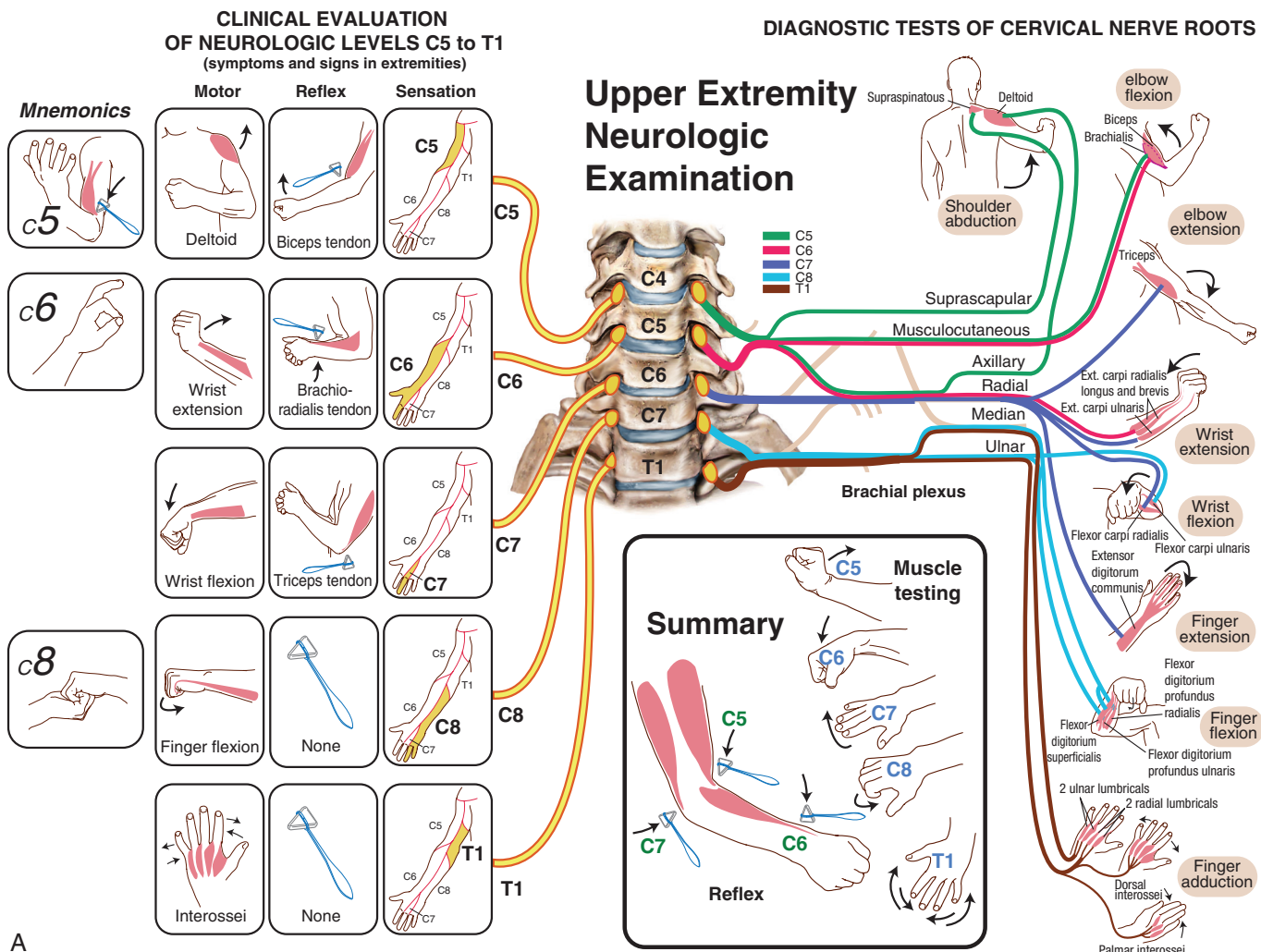
The Spurling test while traditionally used to evaluate for cervical spine radiculopathy, may also be positive in a stinger injury. A positive Spurling test involves the reproduction of paresthesia down the affected extremity upon turning the patients neck toward the side of the suspected lesion, extending the neck, and applying compression of forehead contralaterally. It is thought that the traction applied during a Spurling maneuver may reproduce the mechanism of the stinger injury, thereby also reproducing the symptoms of a stinger. The Spurling test appears to show 30% to 100% sensitivity and 75% to 99% specificity in the literature.<sup>10</sup> Despite the question of variable sensitivity and intra-observer reliability, these examinations may provide useful evidence to an otherwise complicated clinical picture.

The supraspinatus and infraspinatus, along with the biceps and deltoid, appear to be most often involved in stinger injuries.

In cases where a lesion of C5 or C6 nerve roots must be distinguished from an upper trunk lesion, a clinician may test for a long thoracic nerve or dorsal scapular nerve palsy. Involvement of either nerve may result in scapular winging, though long thoracic nerve palsy would result in medial winging during a Serratus wall test, while dorsal scapular nerve lesions exhibit lateral winging and rhomboid weakness.<sup>11</sup>

## IMAGING

Torg et al. first described the Torg ratio in 1986, a metric that is currently used to characterize cervical spinal stenosis.<sup>19</sup> Originally, it was developed to radiographically predict the risk of developing chronic stinger syndrome in the otherwise healthy athlete. On lateral radiograph, the Torg ratio may be calculated by measuring the diameter of the spinal canal (distance from the midpoint of the superior and inferior endplates of the posterior aspect of the vertebral body to the corresponding point on the spinolaminar line). This distance is then divided by the anteroposterior (AP) vertebral body diameter to form the Torg ratio (Fig. 127.2). Initially, a number of studies suggested that a



**Fig. 127.1** (A and B) Examination of the upper extremity for strength, sensation, and reflexes. (From Miller MD, Thompson SR, Hart JA. *Review of Orthopaedics*. Philadelphia: Elsevier; 2012.)

Continued

Lower Extremity  
Neurologic  
Examination

CLINICAL EVALUATION  
OF NEUROLOGIC LEVELS L4 to S1  
(symptoms and signs in extremities)

DIAGNOSTIC TESTS OF  
LUMBOSACRAL NERVE ROOTS

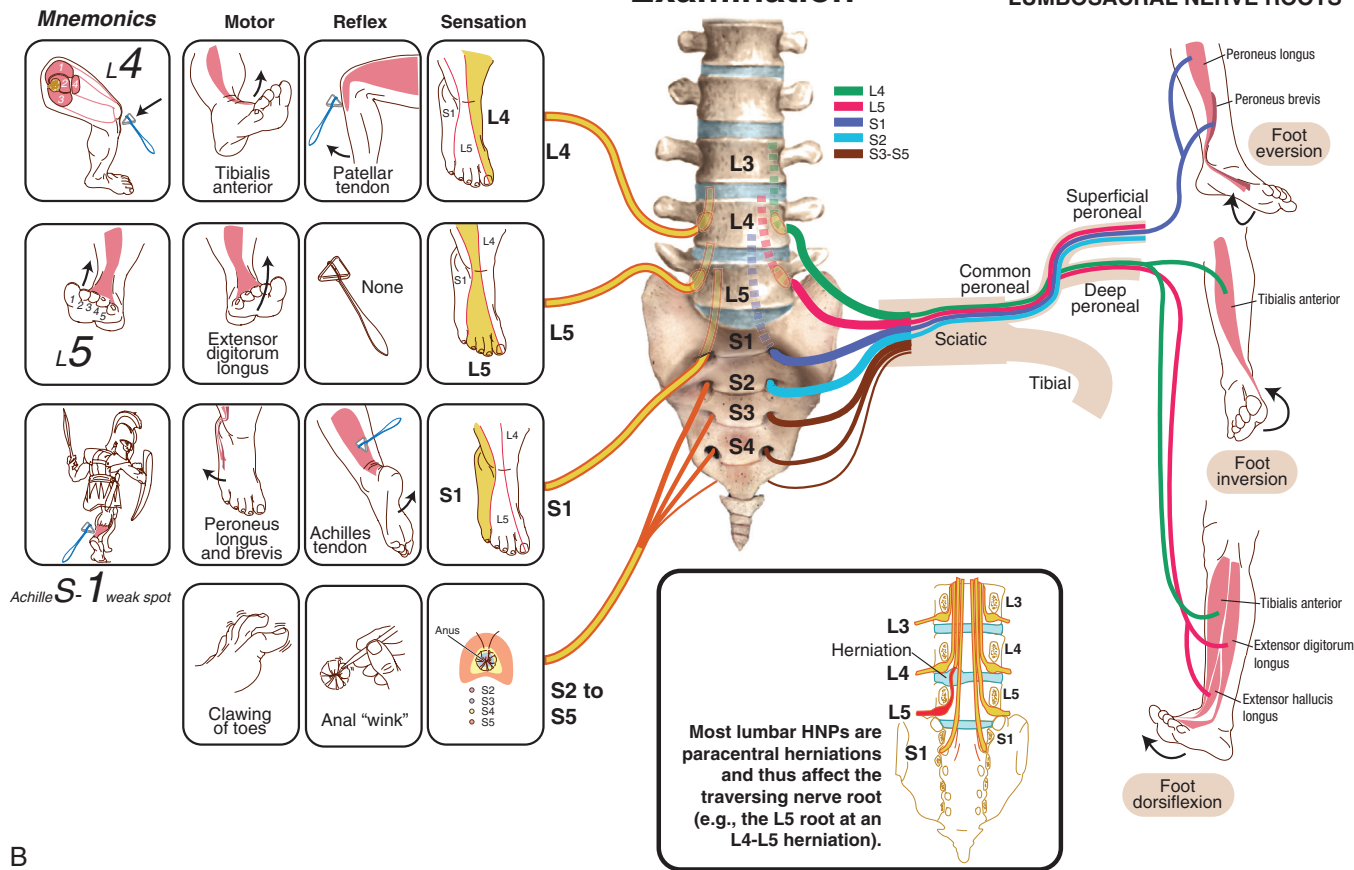


Fig. 127.1, cont'd

TABLE 127.1 Evaluation of Motor Function in Burners and Other Brachial Plexus Injuries

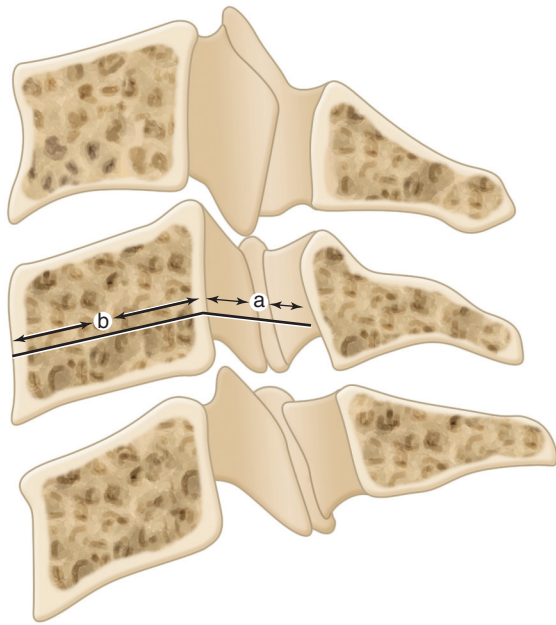
| Muscle                 | Innervation               | Clinical Test         |
|------------------------|---------------------------|-----------------------|
| Deltoid                | Axillary (C5, C6)         | Shoulder abduction    |
| Supraspinatus          | Suprascapular (C5, C6)    | "Full can" abduction  |
| Infraspinatus          | Suprascapular (C5, C6)    | External rotation     |
| Biceps brachii         | Musculocutaneous (C5, C6) | Elbow flexion         |
| Pronator teres         | Median (C6, C7)           | Forearm pronation     |
| Triceps brachii        | Radial (C7, C8)           | Elbow extension       |
| Abductor digiti minimi | Ulnar (C8, T1)            | Fifth digit abduction |

From Kuhlman G, McKeag D. The "burner": a common nerve injury in contact sports. *Am Fam Physician*. 1999;60(7):2.

Torg ratio below 0.7 to 0.8 indicated substantial spinal stenosis and increased risk of a future stinger during contact sports, with lower value corresponding to higher risk. Since the Torg ratio was first described, a number of subsequent studies have found it to have high sensitivity, but relatively poor predictive value and high user dependency.<sup>20,21</sup> Although the Torg ratio is no longer widely used in the setting of stinger injuries, it nevertheless remains an important metric for the classification of cervical stenosis.

More recent literature has explored a novel method of predicting the risk for future stinger injury in the athlete. The mean subaxial cervical space available for the cord (MSCSAC) index—a measurement determined by averaging the difference between sagittal spinal cord diameter and sagittal spinal canal diameter from levels C3 through C6—appears to show some promise in this setting. In theory, any injury mechanism (e.g., hyperextension) that would diminish the MSCSAC would cause undue compression of the underlying cord, predisposing to recurrent stinger injury. In 2009, Presciutti et al. performed a comparison of 103 NFL athletes and 43 age-matched nonathlete controls. After obtaining cervical magnetic resonance imaging (MRI) and radiographs, the Torg ratio and the MSCSAC index were calculated for each subject. They found that MSCSAC was significantly different among athletes with and without chronic stingers, and among athletes and nonathlete controls. Furthermore, an MSCSAC index of less than 4.3 predicted a 13-fold increased risk of chronic stinger development, with 80% sensitivity and 96% specificity.<sup>20</sup> A recent case study describing a Division 1 collegiate football player with chronic stinger syndrome also demonstrated findings consistent with those of Presciutti et al.<sup>21</sup> While the effectiveness of the MSCSAC index as a predictive tool remains promising, further rigorous research in the form





**Fig. 127.2** The ratio method defined by Torg and colleagues uses a cervical spine lateral radiograph to predict an athlete's risk of sustaining a stinger. The Torg ratio is defined as the canal diameter (*a*) divided by the vertebral body diameter (*b*). (Modified from Torg J, Pavlov H, Genuario S, et al. Neurapraxia of the cervical spinal cord with transient quadriplegia. *J Bone Joint Surg.* 1986;68:1354–1370.)

of prospective studies and clinical trials must be undertaken before the use of this method becomes widespread.

While imaging studies are not necessarily required in athletes with a suspected stinger injury, they may be critical for ruling out serious cervical pathology, particularly in those with prolonged, bilateral, or severe symptoms. In the past, lateral flexion and extension radiographs were often used to evaluate the cervical spine. Recently, however, a growing body of literature has supported the replacement of radiographs with computer tomography (CT). In a 2009 prospective study of 1505 patients brought to a level 1 trauma center, traditional radiographs were found to have poor sensitivity for cervical pathology (36%) when compared to CT (100%).<sup>12</sup> When patients were stratified by low, medium, and high risk, based on mechanism of injury, this divide was even more pronounced in low-risk patients. Since then a number of studies, including large randomized trials, have found screening CT to exhibit superior sensitivity, specificity, interobserver, and intraobserver reliability when compared to radiograph, in both obtunded and nonobtunded patients.<sup>13–15</sup> Furthermore, current research suggests that newer techniques of CT imaging, such as adaptive statistical iterative reconstruction (ASiR), may further increase the beneficial value of CT in this setting through reduction in radiation exposure by up to 36%.<sup>16</sup> MRI may be used instead of CT when evaluating stingers in both adult and pediatric populations.<sup>17,18</sup> In general, the acuity and severity of symptomatology should guide the decision to use MRI, which should be based on the individual patient's entire clinical picture.

Electromyography (EMG) has shown some utility in the evaluation of a stinger injury. In addition to differentiating preganglionic and postganglionic lesions, EMG can aid in determining lesion severity.<sup>22</sup> Although the process of wallerian degeneration

begins as early as 24 to 36 hours following injury, nerve segment distal to the lesion may continue to propagate normal signal pattern for up to 2 to 3 weeks. For this reason, EMG is indicated only for those patients with persistent symptoms for a minimum of 2 weeks. EMG findings may be variable, but commonly include normal to increased insertional activity, fibrillations and positive sharp waves at rest, and an incomplete interference pattern.<sup>22,23</sup> Depending on the severity of nerve damage, postganglionic injuries do not appear to preclude spontaneous resolution, whereas preganglionic injuries or nerve root avulsions have little chance for recovery without surgical intervention.

## CLASSIFICATION AND DECISION-MAKING PRINCIPLES

Clinical decision-making, in the context of a stinger injury, necessarily relies on the mechanism of injury. To date, three mechanisms of injury have been explored in the literature. These include direct compression of the Erb point, lateral traction of the brachial plexus, and cervical spine hyperextension (Fig. 127.3). Each of these mechanisms can occur independently, or concomitantly with one another.

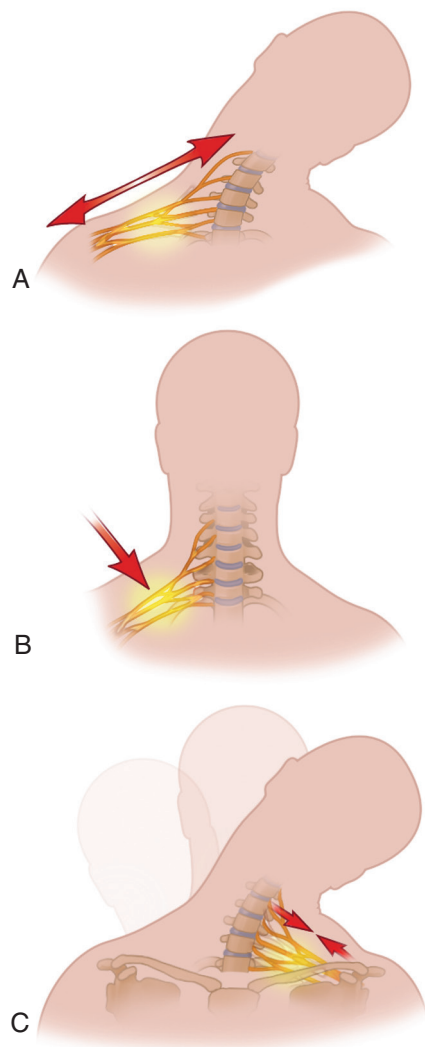
Compression of the superficial nerves at the Erb point is a frequently cited mechanism of stinger injury in the football player. During a direct tackle, the superomedial margin of the scapula can be forced towards a player's shoulder pad, or vice versa. The resultant compressive forces are then transmitted to the upper trunk of the brachial plexus that intervenes between these structures.<sup>24,25</sup> Compressive injury also may occur as a result of any action that forcibly diminishes the diameter of the cervical foramina such as rapid head rotation or pre-existing cervical stenosis.

Brachial plexus traction is a common mechanism of injury in football players, but can also affect wrestlers, rugby players, bicyclists, and skiers. Injury occurs when the athlete head is forcibly bent contralateral to the affected side, and the ipsilateral shoulder is depressed. This can be the result of a direct tackle in the case of contact sports, or landing onto the side of an athlete's helmet following a high-speed fall (cyclists, skiers, etc.).<sup>24</sup>

Hyperextension injuries can occur as a result of a head-on blow to the athlete, often with the force vector facing posteriorly. Here, rapid neck extension compresses the cervical nerve root through brief cervical foramen collapse, resulting in motor and/or sensory deficits. Athletes with degenerative disease of the cervical spine, such as stenosis or disk disease, are more prone to nerve root compression subsequently following a hyperextension event.<sup>25</sup>

Besides the above mentioned three major mechanisms, in young athletes, stinger injuries may be associated with high magnitude acceleration of the head. In one study, Campolettano et al. analyzed 6813 impacts among 34 youth football players. They found that 6.0% of these impacts met the definition of "high magnitude" with head acceleration exceeding 40 g. Overall, the authors found that tackling drills during practice, rather than in-game injury, were responsible for up to 50% of high-magnitude impact exposures among young football players. Furthermore, specific tackling drills, such as the "King of the hill" drill, were strongly associated with head acceleration over 60 g.<sup>26</sup>

Sir Herbert Seddon was among the first to describe injuries to the peripheral nervous system in 1942. Today, his classification scheme remains largely utilized to describe the severity of brachial plexus injuries. Stinger injuries are stratified into grade I, II, and III, with each grade describing varying levels of severity (Table



**Fig. 127.3** Mechanisms of stingers. (A) Traction to the brachial plexus from ipsilateral shoulder depression and contralateral lateral neck flexion. (B) A direct blow to the supraclavicular fossa at the Erb point. (C) Compression of the cervical roots or brachial plexus from ipsilateral lateral flexion and hyperextension. (From Kuhlman G, McKeag D. The “burner”: a common nerve injury in contact sports. *Am Fam Physician*. 1999;60(7): 2035–2040.)

127.2). Grade I injuries represent neuropraxia, in which conduction across a nerve is interrupted despite the presence of full axonal continuity. Although neuropraxias themselves can vary in severity, a key feature is that there is no violation of the epineurium, the perineurium, or the endoneurium of the affected nerve. Wallerian degeneration does not occur, and thus complete recovery is expected within days to weeks of the injury. On EMG, the patient may exhibit conduction block, positive sharp waves, and lack of fibrillation potentials.<sup>2,9,27</sup>

Axonotmesis describes a peripheral nerve injury in which there is axonal disruption, and is indicative of a grade II stinger injury. Grade II stingers are more severe than grade I, but are fortunately less common. Because the epineurium and perineurium are preserved, grade II stingers, axonal regeneration remains possible without surgical intervention. Following injury, nerve conjunction transiently ceases, leading to significant sensorimotor deficits on the affected side. Wallerian degeneration then follows. At 3 weeks post injury, EMG may show some recovery of nerve conduction with fibrillation potentials and positive sharp waves.<sup>2,9</sup> While symptoms may persist for weeks to months, full recovery can take over a year.

Grade III stingers exhibit neurotmesis and are thus the most severe. Neurotmesis involves complete or partial transection of the nerve, including endoneurium, perineurium, epineurium, and any myelinating cells in proximity to the injury. Conduction studies are variable, but can be similar to those found in grade II injuries if the affected nerve is only partially destroyed. Grade III injuries are accompanied by wallerian degeneration, severe sensorimotor, and/or autonomic dysfunction, and require prompt surgical intervention.<sup>27</sup>

TREATMENT OPTIONS

The initial treatment of a stinger injury centers on patient stabilization and removal from play. A thorough history and physical examination should be obtained as described in previous sections of this chapter. A high index of suspicion should be used when evaluating any injured player, as the patient may not immediately alert athletic trainers, especially when symptoms are transient.<sup>24</sup>

Athletes with transient symptoms lasting only seconds, with complete resolution and benign physical exam findings, likely experienced a grade I, mild neuropraxia. Such patients can be managed symptomatically, and may be permitted to immediately RTP at the discretion of the athletic and coaching staff. Any player experiencing symptoms longer than a few seconds likely have a grade I, moderate neuropraxia, and should be removed from play for the remainder of the competition.

| TABLE 127.2 Classification of Stinger Injury Defined by Nerve Injury, Electromyographic Findings, and Prognosis for Recovery |              |                                  |                                              |
|------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------|----------------------------------------------|
| Grade                                                                                                                        | Nerve Injury | Electromyographic Findings       | Prognosis                                    |
| 1                                                                                                                            | Neurapraxia  | Normal                           | Most resolve within minutes                  |
| 2                                                                                                                            | Axonotmesis  | Positive waves with fibrillation | Recovery in 12–18 months                     |
| 3                                                                                                                            | Neurotmesis  | Acute denervation                | Variable; possible complete loss of function |

Physical exam should be performed at the time of injury, at the end of play, and at 24 hours after injury for these individuals. The patient should also undergo cervical radiography, particularly if strength, ROM, and sensation has not returned to baseline. Treatment grade I, severe neuropraxia incorporates all of the above steps, and additionally includes repeated physical examination of the patient up to 2 weeks following injury.<sup>24,28</sup> Additional imaging such as MRI or CT may be considered on a per-patient basis.

Serial examination and cervical imaging are also warranted in patients with grade II stingers. These patients should be closely followed for up to 2 weeks after initial injury, in a similar fashion to those with severe neuropraxia. Cervical imaging options include standard AP, lateral, and odontoid view radiographs, as well as MRI; CT also may be used if the patient develops symptoms suspicious of a cervical fracture.<sup>24,28</sup>

Patients with grade III injury will exhibit severe weakness and sensorimotor dysfunction lasting more than 2 to 3 weeks. For these individuals, nerve conduction studies, such as EMG, are valuable for establishing nerve function and monitoring

rehabilitation. MRI and conduction study findings should guide clinical decision-making, and surgical consultation is indicated for severe irreparable nerve damage.<sup>24,28,29</sup>

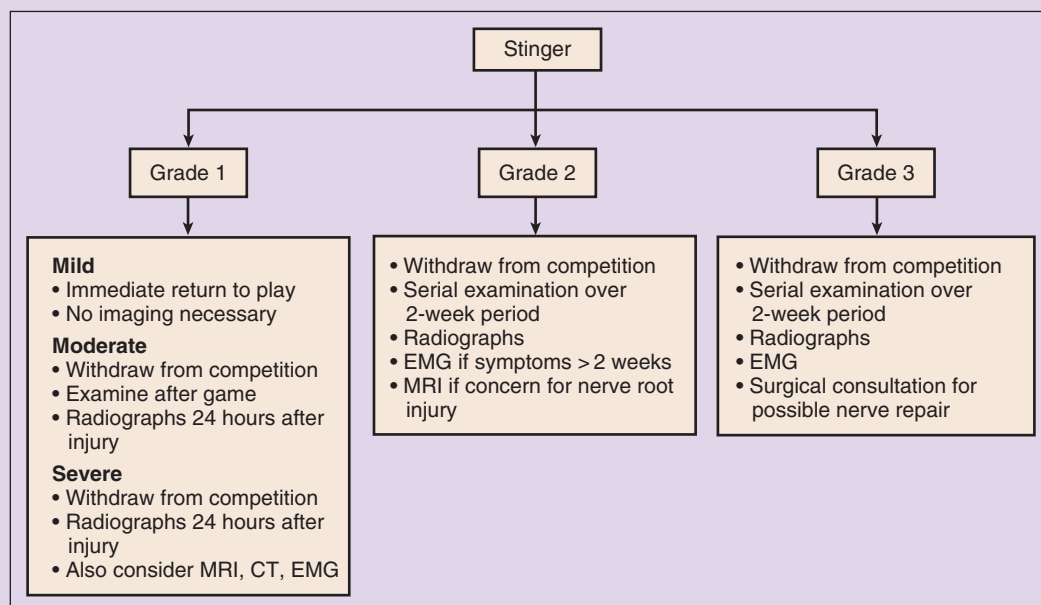
Symptomatic treatment of nerve injury-related pain is indicated in all patients regardless of injury grade. First-line treatment includes the application of ice and the use of nonsteroidal antiinflammatory drugs (NSAIDs) to address local inflammation. Physical therapy is of value to those with symptoms that limit ranging of the cervical spine and the affected extremity. In this situation, an emphasis should be placed on regaining active and passive neck ROM, and improving the strength of musculature surrounding the site of injury. While the use of high-dose steroids remains contraindicated in patients with peripheral nerve injuries, a short course of low-dose oral steroids in patients with severe symptoms is generally accepted among most physicians.<sup>28,29</sup> Similarly, cortisone injection near the site of nerve injury also may provide some symptomatic relief, though there are currently no concrete guidelines surrounding this. Muscle relaxants may be judiciously used in the patient with pain secondary to spasm of the musculature surrounding the injury.<sup>30</sup>



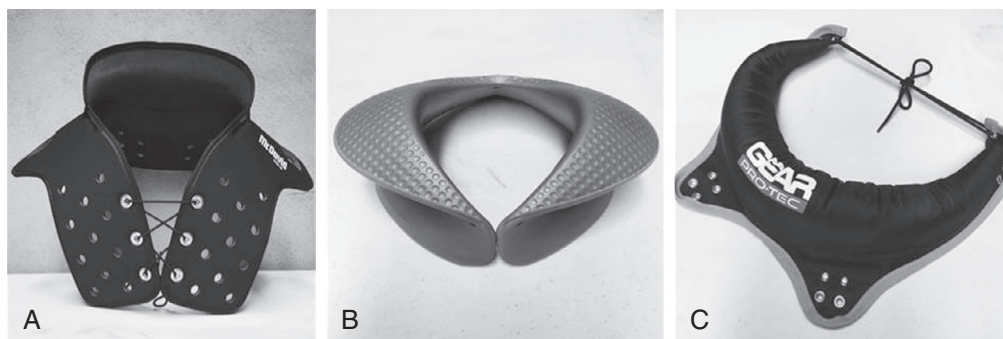
### Authors' Preferred Technique

Given the prevalence of stingers, the potential for life-altering complications and comorbid injuries, and the high propensity of underreporting observed throughout the literature, the authors advocate for the evaluation of all athletes exhibiting neurologic symptoms following in-game trauma. Any athlete sustaining symptoms that persist during sideline evaluation should be removed from competition and reevaluated after the game and 24 hours after injury. Unstable cervical injury should be managed according to local protocol; usually cervical CT should be obtained in patients exhibiting signs suggestive of fracture such as focal cervical tenderness. We find that initial CT imaging provides definitive information regarding cervical injury, and is generally appropriate for most injury patterns we encounter. Radiographs are an appropriate alternative if CT imaging is not

available. If symptoms persist after 3 days, an MRI should be obtained. Serial imaging studies and exams are performed for 2–3 weeks following injury, until symptoms show improvement. If symptoms do not resolve at this time, EMG studies are obtained. We do not routinely use cortisone injections, oral steroids, or muscle relaxants, but do encourage rest, ice, the use of NSAIDs, and physical therapy when appropriate. Patients may be allowed to RTP following (1) complete resolution of pain, numbness, and tingling, (2) achievement of baseline muscle strength and ROM, and (3) subjective feeling of “readiness” to return to sport (Fig. 127.4). The patient should be cautioned that recurrence of injury is common, and should be encouraged to use protective equipment in the future.



**Fig. 127.4** Management of stingers based on grade of injury. CT, Computed tomography; EMG, electromyography; MRI, magnetic resonance imaging.



**Fig. 127.5** Examples of protective and preventative equipment. (A) Anterior view of the Cowboy Collar, typically worn under the athlete's shoulder pads. (B) The Kerr Collar, a protective collar that reduces axial load and prevents neck hyperextension. (C) The Cervical Neck Roll, worn under protective padding, also prevents excessive neck motion.

## RETURN TO PLAY

The exact criteria for RTP among athletes with a stinger injury remains controversial. Still, the vast majority of physicians agree that complete resolution of symptoms, the return of preinjury motor strength and ROM, and the ruling out of other serious pathology on imaging, are all prerequisites to RTP.<sup>24,31</sup> Typically, fulfillment of these criteria within a few minutes of injury, in a patient that has had no prior stingers, is sufficient for immediate RTP. In the past, Weinstein has recommended that athletes with moderate fibrillation potentials on EMG, and function deficits 2 weeks after injury should not be allowed to RTP, though the relationship between EMG results and recovery is not entirely clear.<sup>32</sup> Cantu has suggested that a history of 3 or more previous stingers should be a relative contraindication to RTP. Kepler has added that patients with recurrent stingers, particularly within the same season, should be radiographically cleared to RTP following each episode. Following two or more stinger injuries, experts suggest that the patient be counseled to review proper blocking and tackling technique to minimize the risk of future injury, prior to return.<sup>28</sup> Additionally, the clinician should facilitate a thorough conversation about the risk of future recurrence, and encourage the use of protective equipment, such as high shoulder pads or a cervical roll, prior to return to sport.<sup>29</sup>

## SEQUELA

Severe complications are rarely seen following a stinger. Over 90% of patients experience symptomatic improvement within 24 hours of injury. In those with prolonged symptoms, the most life-altering complication is permanent neurologic dysfunction. This is most commonly seen in individuals with grade III injury and complete nerve transection. Without surgical intervention and subsequent rehabilitation, these individuals rarely recover. A more common complication of a stinger injury is recurrence. Indeed, 50% to 60% of those with a previous stinger will experience one or more later episodes.<sup>33</sup> Levitz et al. were among the first to describe the chronic stinger syndrome, and suggested that cervical foraminal stenosis and degenerative disk disease

may be associated with an increased risk of future recurrence.<sup>33</sup> As mentioned in previous sections of this text, the Torg ratio has historically been used as a quantitative metric to predict the risk for the development of chronic stinger syndrome.<sup>19</sup> More recently, the MSCSAC index has been proposed to better serve this purpose, with promising evidence supporting its predictive value.<sup>20,21</sup>

## FUTURE CONSIDERATIONS

The prevention of initial injury and injury recurrence remains a target of future research. Recent literature has suggested that the use of the "Kerr Collar" minimizes head acceleration and force transmission better than the use of the "Cowboy" or "Bullock Collars" do, although all of these collars have significant protective value.<sup>34</sup> (Fig. 127.5). However, Concannon et al. have cautioned that equipment that excessively limits head extension may theoretically increase the risk for other cervical injury.<sup>35</sup> This remains an area of active research, and further biomechanical studies may best elucidate the optimal head and neck positioning to prevent stingers. The MSCSAC index is a recently devised metric used to aid in prediction of future stinger injury recurrence. While it shows significant promise, it has not been extensively studied in youth and collegiate populations. Furthermore, future research must aim to establish baseline MSCSAC index values in athletes at the professional, semiprofessional, and recreational levels.<sup>20,36,37</sup>

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

## SELECTED READINGS

### Citation:

Presciutti SM, Deluca P, Marchetto P, et al. Mean subaxial space available for the cord index as a novel method of measuring cervical spine geometry to predict the chronic stinger syndrome in American football players. *J Neurosurg Spine*. 2009;11(3):264–271.

### Level of Evidence:

II



**Summary:**

Presciutti et al. propose and evaluate the use of the mean subaxial space available for the cord (MSCSAC) index as an alternative quantitative parameter to the Torg ratio for predicting the risk of recurrent stingers. The authors retrospectively reviewed cervical magnetic resonance imaging from 103 male NFL athletes and 43 age-matched controls. Torg ratio and MSCSAC index were calculated, and receiver operator curves were obtained for each of the modalities. An MSCSAC index value of 5.0 mm produced a sensitivity of 80% and a negative likelihood ratio of 0.23 for predicting chronic stingers. The authors concluded that MSCSAC index measurement may be an appropriate screening test in at-risk patients.

**Citation:**

Huang P, Anissipour A, McGee W, et al. Return-to-play recommendations after cervical, thoracic, and lumbar spine injuries. *Sports Health*. 2016;8(1):19–25.

**Level of Evidence:**

IV

**Summary:**

Huang et al. provide a comprehensive review and discussion of return to play guidelines for a wide range of cervical injuries. Their work provides useful references to both the current literature and expert opinion pertaining to the management of

athletes with stingers and other trauma-mediated injury. They conclude that, at a minimum, individuals with cervical injuries should have complete symptomatic relief and return to baseline strength and range of movement prior to return to play. Finally, they cover injury-specific nuances in management of the injured athlete

**Citation:**

Green J, Zuckerman SL, Dalton SL, et al. A 6-year surveillance study of “Stingers” in NCAA American Football. *Res Sports Med*. 2016;25(1):26–36.

**Level of Evidence:**

II

**Summary:**

In this retrospective review of prospectively collected data, Green et al. report on the prevalence of stingers reported among 57 NCAA football teams over the course of 6 years. They found an injury rate of 2.04 per 10,000 athlete exposures. Stingers were reported primarily in the preseason (80.3%) and 18.8% of injuries were recurrent. The most high-risk positions for injury were defensive ends/linebackers (25.8%) and offensive linemen (23.6%). The authors conclude that stingers may be less common than previously reported in the literature within the collegiate football population.

## REFERENCES

- Chrisman D, Snook G, Stanitis J, et al. Lateral-flexion neck injuries in athletic competition. *JAMA*. 1965;192(7):613–615.
- Tosti R, Rossy W, Sanchez A, et al. Burners, stingers, and other brachial plexus injuries in the contact athlete. *Oper Tech Sports Med*. 2016;24(4):273–277. doi:10.1053/j.otsm.2016.09.006.
- Naden CM. Brachial plexopathy. *Curr Sports Med Rep*. 2017;16(3):121.
- Paulus S, Kennedy D. Return to play considerations for cervical spine injuries in athletes. *Phys Med Rehabil Clin N Am*. 2014;25(4):723–733. doi:10.1016/j.pmr.2014.06.005.
- Charbonneau R, McVeigh S, Thompson K. Brachial neuropraxia in Canadian Atlantic University sport football players. *Clin J Sport Med*. 2012;22(6):472–477. doi:10.1097/jsm.0b013e3182699ed5.
- Green J, Zuckerman S, Dalton S, et al. A 6-year surveillance study of “Stingers” in NCAA American Football. *Research in Sports Medicine*. 2016;25(1):26–36. doi:10.1080/15438627.2016.1258642.
- Kawasaki T, Ota C, Yoneda T, et al. Incidence of stingers in young rugby players. *Am J Sports Med*. 2015;43(11):2809–2815. doi:10.1177/0363546515597678.
- Herman M. Cervical spine injuries in the pediatric adolescent athlete. *Instr Course Lect*. 2006;55:641–646.
- Aval SM, Durand P, Shankwiler JA. Neurovascular injuries to the athlete’s shoulder: part I. *J Am Acad Orthop Surg*. 2007;15(4):249–256. doi:10.5435/00124635-200704000-00008.
- Rubinstein SM, Pool JJM, Tulder MWV, et al. A systematic review of the diagnostic accuracy of provocative tests of the neck for diagnosing cervical radiculopathy. *Eur Spine J*. 2007;16(3):307–319. doi:10.1007/s00586-006-0225-6.
- Thatte M, Babhulkar S, Hiremath A. Brachial plexus injury in adults: diagnosis and surgical treatment strategies. *Ann Indian Acad Neurol*. 2013;16(1):26. doi:10.4103/0972-2327.107686.
- Bailitz J, Starr F, Beecroft M, et al. CT should replace three-view radiographs as the initial screening test in patients at high, moderate, and low risk for blunt cervical spine injury: a prospective comparison. *J Trauma*. 2009;66(6):1605–1609.
- Middendorp JJV, Cheung I, Dalzell K, et al. Detecting facet joint and lateral mass injuries of the subaxial cervical spine in major trauma patients. *Asian Spine J*. 2015;9(3):327. doi:10.4184/asj.2015.9.3.327.
- Inaba K, Byerly S, Bush LD, et al. Cervical spinal clearance. *J Trauma Acute Care Surg*. 2016;81(6):1122–1130. doi:10.1097/ta.0000000000001194.
- Patel MB, Humble SS, Cullinane DC, et al. Cervical spine collar clearance in the obtunded adult blunt trauma patient. *J Trauma Acute Care Surg*. 2015;78(2):430–441. doi:10.1097/ta.0000000000000503.
- Patro SN, Chakraborty S, Sheikh A. The use of adaptive statistical iterative reconstruction (ASiR) technique in evaluation of patients with cervical spine trauma: impact on radiation dose reduction and image quality. *Br J Radiol*. 2016;89(1060):20150082. doi:10.1259/bjr.20150082.
- Henry M, Riesenburger RI, Kryzanski J, et al. A retrospective comparison of CT and MRI in detecting pediatric cervical spine injury. *Childs Nerv Syst*. 2013;29(8):1333–1338. doi:10.1007/s00381-013-2092-x.
- Sola R, Christmas A, Thomas B, et al. Do not waste your time: straight to magnetic resonance imaging for pediatric burners and stingers. *Am J Emerg Med*. 2016;34(8):1442–1445. doi:10.1016/j.ajem.2016.04.020.
- Torg J, Pavlov H, Genuario S, et al. Neurapraxia of the cervical spinal cord with transient quadriplegia. *J Bone Joint Surg Am*. 1986;68:1354–1370.
- Presciutti SM, Deluca P, Marchetto P, et al. Mean subaxial space available for the cord index as a novel method of measuring cervical spine geometry to predict the chronic stinger syndrome in American football players. *J Neurosurg Spine*. 2009;11(3):264–271. doi:10.3171/2009.3.spine08642.
- Greenberg J, Leung D, Kendall J. Predicting chronic stinger syndrome using the mean subaxial space available for the cord index. *Sports Health*. 2011;3(3):264–267. doi:10.1177/1941738111403866.
- Saliba S, Saliba EN, Pugh KF, et al. Rehabilitation considerations of a brachial plexus injury with complete avulsion of C5 and C6 nerve roots in a college football player: a case study. *Sports Health*. 2009;1(5):370–375. doi:10.1177/1941738109343544.
- Rezende M, Silva G, Paula E, et al. What has changed in brachial plexus surgery? *Clinics*. 2013;68(3):411–418. doi:10.6061/clinics/2013(03)r02.
- Huang P, Anissipour A, McGee W, et al. Return-to-play recommendations after cervical, thoracic, and lumbar spine injuries: a comprehensive review. *Sports Health*. 2016;8(1):19–25. doi:10.1177/1941738115610753.
- Rowson S, Brolinson G, Goforth M, et al. Linear and angular head acceleration measurements in collegiate football. *J Biomech Eng*. 2009;131(6):061016. doi:10.1115/1.3130454.
- Campolettano ET, Rowson S, Duma SM. Drill-specific head impact exposure in youth football practice. *J Neurosurg Pediatr*. 2016;18(5):536–541. doi:10.3171/2016.5.peds1696.
- Chao S, Pacella MJ, Torg JS. The pathomechanics, pathophysiology and prevention of cervical spinal cord and brachial plexus injuries in athletics. *Sports Med*. 2010;40(1):59–75. doi:10.2165/11319650-000000000-00000.
- Cantu RC, Li YM, Abdulhamid M, et al. Return to play after cervical spine injury in sports. *Curr Sports Med Rep*. 2013;12(1):14–17. doi:10.1249/jsr.0b013e31827dc1fb.
- Kepler CK, Vaccaro AR. Injuries and abnormalities of the cervical spine and return to play criteria. *Clin Sports Med*. 2012;31(3):499–508. doi:10.1016/j.csm.2012.03.005.
- Olivo R, Tsao B. Peripheral nerve injuries in sport. *Neurol Clin*. 2017;35(3):559–572.
- Rosenthal B, Boody B, Hsu W. Return to play for athletes. *Neurosurg Clin N Am*. 2017;28(1):163–171. doi:10.1016/j.nec.2016.08.003.
- Weinstein S. Assessment and rehabilitation of an athlete with a “stinger”: a model for the management of noncatastrophic athletic cervical spine injury. *Clin Sports Med*. 1998;17:127–135.
- Levitz CL, Reilly PJ, Torg JS. The pathomechanics of chronic, recurrent cervical nerve root neurapraxia. *Am J Sports Med*. 1997;25(1):73–76. doi:10.1177/036354659702500114.

34. Rowson S, Mcneely DE, Brolinson PG, et al. Biomechanical analysis of football neck collars. *Clin J Sport Med*. 2008;18(4):316–321. doi:10.1097/jsm.0b013e31817f016a.
35. Concannon LG, Harrast MA, Herring SA. Radiating upper limb pain in the contact sport athlete. *Curr Sports Med Rep*. 2012;11(1):28–34. doi:10.1249/jsr.0b013e318240dc3f.
36. Zaremski JL, Horodyski M, Herman DC. Recurrent stingers in an adolescent football player: dilemmas of return to play. A case report and review of the literature. *Res Sports Med*. 2017;25(3):384–390. doi:10.1080/15438627.2017.1314297.
37. Olivo R, Tsao B. Peripheral nerve injuries in sport. *Neurol Clin*. 2017;35(3):559–572.